Chapter Review

Key Terms

coordinate A number paired with a point on a number line is called the coordinate of the point.
counting numbers The counting numbers are the numbers 1, 2, 3, ...
difference The difference is the result of subtracting two or more numbers.
dividend When dividing two numbers, the dividend is the number being divided.
divisor When dividing two numbers, the divisor is the number dividing the dividend.
number line A number line is used to visualize numbers. The numbers on the number line get larger as they go from left to right, and smaller as they go from right to left.
origin The origin is the point labeled 0 on a number line.
place value system Our number system is called a place value system because the value of a digit depends on its position, or place, in a number.
product The product is the result of multiplying two or more numbers.
quotient The quotient is the result of dividing two numbers.
rounding The process of approximating a number is called rounding.
sum The sum is the result of adding two or more numbers.
whole numbers The whole numbers are the numbers 0, 1, 2, 3, ...

Key Concepts

1.1 Introduction to Whole Numbers

Place Value														
Trillions			Billions			Millions			Thousands			Ones		
Hundred trillions	Ten trillions	Trillions	Hundred billions	Ten billions	Billions	Hundred millions	Ten millions	Millions	Hundred thousands	Ten thousands	Thousands	Hundreds	Tens	Ones
								5	2	7	8	1	9	4

Figure 1.16

- **Name a whole number in words.**
  Step 1. Starting at the digit on the left, name the number in each period, followed by the period name. Do not include the period name for the ones.
  Step 2. Use commas in the number to separate the periods.
- **Use place value to write a whole number.**
  Step 1. Identify the words that indicate periods. (Remember the ones period is never named.)
  Step 2. Draw three blanks to indicate the number of places needed in each period.
  Step 3. Name the number in each period and place the digits in the correct place value position.
- **Round a whole number to a specific place value.**
  Step 1. Locate the given place value. All digits to the left of that place value do not change unless the given place value is a 9, in which case it may. (See Step 3.).
  Step 2. Underline the digit to the right of the given place value.
  Step 3. Determine if this digit is greater than or equal to 5. If yes—add 1 to the digit in the given place value. If that digit is 9, replace it with 0 and add 1 to the digit immediately to its left. If that digit is also a 9, repeat. If no—do not change the digit in the given place value.
  Step 4. Replace all digits to the right of the given place value with zeros.

1.2 Add Whole Numbers

- **Addition Notation** To describe addition, we can use symbols and words.

Operation	Notation	Expression	Read as	Result
Addition	$+$	$3+4$	three plus four	the sum of 3 and 4

- **Identity Property of Addition**
  - The sum of any number a and 0 is the number. $a+0=a\ 0+a=a$
- **Commutative Property of Addition**
  - Changing the order of the addends a and b does not change their sum. $a+b=b+a$.
- **Add whole numbers.**
  Step 1. Write the numbers so each place value lines up vertically.
  Step 2. Add the digits in each place value. Work from right to left starting with the ones place. If a sum in a place value is more than 9, carry to the next place value.
  Step 3. Continue adding each place value from right to left, adding each place value and carrying if needed.

1.3 Subtract Whole Numbers

Operation	Notation	Expression	Read as	Result
Subtraction	$-$	$7-3$	seven minus three	the difference of 7 and 3

- **Subtract whole numbers.**
  Step 1. Write the numbers so each place value lines up vertically.
  Step 2. Subtract the digits in each place value. Work from right to left starting with the ones place. If the digit on top is less than the digit below, borrow as needed.
  Step 3. Continue subtracting each place value from right to left, borrowing if needed.
  Step 4. Check by adding.

1.4 Multiply Whole Numbers

Operation	Notation	Expression	Read as	Result
Multiplication	$\times$ $\cdot$ $()$	3×8 $3\cdot 8$ $3(8)$	three times eight	the product of 3 and 8

- **Multiplication Property of Zero**
  - The product of any number and 0 is 0.
    $a\cdot 0=0$
    $0\cdot a=0$
- **Identity Property of Multiplication**
  - The product of any number and 1 is the number.
    $1\cdot a=a$
    $a\cdot 1=a$
- **Commutative Property of Multiplication**
  - Changing the order of the factors does not change their product.
    $a\cdot b=b\cdot a$
- **Multiply two whole numbers to find the product.**
  Step 1. Write the numbers so each place value lines up vertically.
  Step 2. Multiply the digits in each place value.
  Step 3. Work from right to left, starting with the ones place in the bottom number.
  Step 4. Multiply the bottom number by the ones digit in the top number, then by the tens digit, and so on.
  Step 5. If a product in a place value is more than 9, carry to the next place value.